IDS (Basics)

Network Surveillance, Snort Methodology, and Rule Engineering

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Intrusion Detection Systems (IDS) provide a vital layer of digital surveillance, acting as the "security cameras" of a network. While firewalls function as gatekeepers to permit or deny connections at the perimeter, an IDS monitors active traffic to detect malicious activity that has already bypassed initial defenses. This module explores the deployment of **Host-based (HIDS)** and **Network-based (NIDS)** systems, alongside detection strategies such as **Signature-based** and **Anomaly-based** logic.

We focus specifically on **Snort**, the industry-standard open-source IDS, detailing its operational modes and the precise syntax required to engineer custom detection rules. By mastering these concepts, security professionals can effectively monitor environments for zero-day threats and historical breaches through PCAP analysis.

This **documentation** and **learning** approach is strongly inspired by the structured, hands-on content provided by **TryHackMe**, which significantly contributed to practical understanding of "IDS" concepts.

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1. What is an IDS?

An **IDS** is a security solution that monitors a network or system for malicious activities or policy violations.

- **Primary Function:** To **Detect** and **Alert**. Unlike an IPS (Intrusion Prevention System), a standard IDS does not stop the threat; it notifies administrators to take action.
- **Firewall vs. IDS:**
  - **Firewall:** A gatekeeper that checks traffic at the point of entry/exit.
  - **IDS:** A surveillance system that monitors behavior once the user is already "inside" the network.

2. Types of IDS

Deployment Modes

Mode	Scope	Description
HIDS	Single Host	Installed on a specific device (e.g., a server) to monitor internal logs and files.
NIDS	Entire Network	Monitors traffic for all devices on a network segment. Requires **Promiscuous Mode**.

Detection Modes

- **Signature-Based:** Matches traffic against a database of known attack patterns (**Signatures**). It is fast but cannot detect **Zero-Day** attacks.
- **Anomaly-Based:** Establishes a **Baseline** of normal behavior and alerts on any deviation. It can catch new threats but often produces **False Positives**.
- **Hybrid:** Combines both methods to maximize detection accuracy.

3. Snort IDS: Operational Modes

Snort is a powerful, rule-based IDS. It operates in three distinct modes depending on the administrator's needs:

1. **Packet Sniffer Mode:** Simply reads and displays packets on the console (Troubleshooting).
2. **Packet Logging Mode:** Records packets to the disk in **PCAP** format (Forensics).
3. **NIDS Mode:** Analyzes traffic in real-time against rule files and generates alerts (Primary Security Function).

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4. Snort Rule Engineering

Snort uses a specific syntax to define what constitutes a threat. A rule consists of the **Rule Header** (Action, Protocol, IPs, Ports) and **Rule Options** (Metadata).

Rule Breakdown Example

alert icmp any any -> $HOME_NET any (msg:"Ping Detected"; sid:10001; rev:1;)

Component	Value	Description
Action	alert	The step to take (e.g., alert, log, pass).
Protocol	icmp	The network protocol to monitor.
Source IP/Port	any any	Traffic originating from any IP and any port.
Direction	->	Indicates the flow of traffic.
Dest IP/Port	$HOME_NET any	Reaching the internal network variable on any port.
Metadata	msg, sid, rev	The alert message, unique ID, and version number.

Command Section: Running Snort

Task	Command
Monitor Real-time	sudo snort -q -l /var/log/snort -i eth0 -A console -c /etc/snort/snort.conf
Analyze PCAP File	sudo snort -q -r capture.pcap -A console -c /etc/snort/snort.conf

5. Practical Lab (Task 5)

I researched the provided network capture and **found** several indicators of a targeted attack. Below is how **I solved** the forensic analysis:

- **Work Done: I practiced** navigating to the /etc/snort directory to locate the suspicious traffic file Intro_to_IDS.pcap.
- **Solution: I practiced** running Snort in analysis mode using the command sudo snort -r Intro_to_IDS.pcap -c /etc/snort/snort.conf. **I founded** that the engine successfully identified multiple alerts based on the custom rules provided.
- **Intrusion Analysis: I researched** the console output and **found** that the attacker's IP address was **10.10.160.187**.
- **Detection Details: I solved** the identification of the attack type by checking the **sid** (Signature ID) in the logs. **I founded** that the specific rule detecting the SSH attempt had an **sid of 1000001**.
- **Result: I founded** a second alert message in the PCAP file labeled "ICMP Packet found," indicating the attacker performed network discovery before the exploit attempt.

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Conclusion and Cheatsheet

Conclusion

An IDS is a fundamental component of a "defense-in-depth" strategy. While firewalls provide the first line of defense, the IDS ensures visibility into internal movements and complex attack patterns. **Snort** remains a critical tool for any SOC analyst, offering the flexibility to create custom rules that adapt to an organization's specific threat landscape. Whether analyzing real-time traffic or historical **PCAP** files, the ability to interpret and write Snort rules is essential for timely incident detection.

Fast Revision Cheatsheet

- **IDS Purpose:** Detection and Notification (No Prevention).
- **HIDS:** Host-level visibility.
- **NIDS:** Network-wide visibility (requires Promiscuous Mode).
- **Signature-Based:** Best for known threats; uses a database.
- **Anomaly-Based:** Best for unknown/Zero-Day threats; uses a baseline.
- **Snort Main Config:** /etc/snort/snort.conf.
- **Custom Rules Path:** /etc/snort/rules/local.rules.
- **sid:** Unique Signature ID (Custom rules usually start at 1,000,001).
- **rev:** Rule revision number (increments with changes).

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